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WAVEGUIDE WITH TRANSITIONAL PORTION AND METHOD OF FABRICATING SAME

Abstract

A waveguide has a first and second portions, and a transitional portion with a first end joining the first portion and a second end joining the second portion. The first portion has a first thickness that is smaller than a second thickness of the second portion. The transitional portion has a thickness that gradually increases from the first thickness at its first end to the second thickness at its second end. In a fabrication method employing chemical-mechanical polishing (CMP), first and second CMP control structures are disposed on opposite sides of the waveguide. Spaces between the waveguide and the CMP control structures are filled with cladding material. CMP is performed to reduce a thickness of the waveguide. The CMP control structures control the CMP of the waveguide to form the transitional portion of the waveguide having the gradually increasing thickness.

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Background/Summary

BACKGROUND

[0001] The following relates to optical waveguides, methods of making optical waveguides, optical waveguide couplings, and the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. 1 diagrammatically illustrates a top view of an optical structure including a waveguide.

[0004] FIG. 2 diagrammatically illustrates a cut view of the waveguide of the optical structure of FIG. 1 taken along cut C0-C0 indicated in FIG. 1.

[0005] FIG. 3 diagrammatically illustrates the top view of the optical structure of FIGS. 1 and 2, with certain dimensions labeled and with cut planes indicated for a cut C1-C1, a cut C2-C2, and a cut C3-C3.

[0006] FIG. 4 diagrammatically illustrates the cut view of the waveguide of FIG. 2, with certain dimensions labeled, and with the cut planes indicated for the same cuts C1-C1, C2-C2, and C3-C3 indicated in FIG. 3.

[0007] FIGS. 5A, 5B, 5C, 5D, 5E, 5F, 5G, and 5H diagrammatically illustrate cut C3-C3 indicated in FIGS. 3 and 4 at successive steps of a waveguide fabrication process.

[0008] FIG. 6 diagrammatically illustrates two cut views taken along cut C1-C1 indicated in FIGS. 3 and 4, with the left-hand cut view depicting the structure before CMP and the right-hand cut view depicting the structure after CMP.

[0009] FIG. 7 diagrammatically illustrates two cut views taken along cut C2-C2 indicated in FIGS. 3 and 4, with the left-hand cut view depicting the structure before CMP and the right-hand cut view depicting the structure after CMP.

[0010] FIG. 8 diagrammatically illustrates two cut views taken along cut C3-C3 indicated in FIGS. 3 and 4, with the left-hand cut view depicting the structure before CMP and the right-hand cut view depicting the structure after CMP.

[0011] FIG. 9 diagrammatically illustrates a cut view of an optical waveguide having a portion with a continuously varying waveguide height, delivering light to an optical detector.

DETAILED DESCRIPTION

[0012] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and

second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0013] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated **90** degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0014] Silicon waveguides are used as silicon photonic components, integrated electrical/optical circuits, or the like. A silicon waveguide typically employs silicon as the higher refractive index core that carries the light, and silicon dioxide as a cladding having lower refractive index. Silicon waveguides are manufactured using principally silicon and silicon dioxide, and are well suited for integration with existing silicon-based microelectronic and integrated circuit (IC) technologies.

[0015] In certain applications, it may be useful for the silicon waveguide to have portions with different thicknesses (i.e., different heights). For example, a light input coupling waveguide portion may have a reduced height that is optimized to match with the size of a light emitter, or a photodetector coupling may have a larger height optimized to match a photodetector, as two nonlimiting illustrative examples. Having waveguide portions with different heights increases the manufacturing complexity, as each different thickness (i.e., height) in the silicon waveguide entails an additional photolithographic patterning and etching process sequence.

[0016] A further issue is that each change in height of a portion of a waveguide introduces an abrupt step in the height of the waveguide. Light reflection occurs at the height step, and the reflected light constitutes optical loss. The light reflection can also produce undesirable destructive optical interference. In the case of polychromatic light, the amount of reflection and constructive or destructive interference can be wavelength-dependent, which can produce undesirable wavelength dependence on the transmission efficiency of the waveguide.

[0017] In the following, improved waveguides are disclosed which advantageously provide a gradual transition between waveguide portions of different thicknesses (i.e., different heights), without an abrupt step. The gradual transition substantially reduces or eliminates light reflection at the transition. Also disclosed herein are methods of manufacturing such improved waveguides, which advantageously achieve the gradual transition from one thickness to another without multiple photolithographic patterning and etching process sequences. The disclosed manufacturing processes utilize a chemical-mechanical polishing (CMP) process controlled by a CMP control structure to form a transitional portion of the waveguide having a gradually changing thickness.

[0018] With reference to FIGS. **1** and **2**, an optical structure including a waveguide **10** is shown by way of a top view in FIG. **1**; and FIG. **2** shows a cut view of the waveguide **10** of the optical structure of FIG. **1** taken along cut **C0-C0** indicated in FIG. **1**. Without loss of generality, a Cartesian coordinate set of directions *x*, *y*, and *z* are indicated in FIGS. **1** and **2** (as well as in a number of other drawings herein). In this system, the plane of the semiconductor wafer or substrate (not shown) on which the optical structure is fabricated is denoted as an *x-y* plane, and the *z*-direction is transverse to the *x-y* plane of the wafer. Note that the depicted Cartesian coordinate set of directions *x*, *y*, *z* does not indicate any particular origin (i.e., no origin where *x*=0, *y*=0, and *z*=0 is indicated); rather, the depicted Cartesian coordinate set of directions indicate three mutually orthogonal *x*-, *y*-, and *z*-directions. In this coordinate system, a layer disposed on the wafer or substrate has a thickness in the *z*-direction. The thickness may also be referred to as a height, and the terms “thickness” and “height” are used interchangeably herein, and both refer to the direction perpendicular to the principal surface of the wafer or substrate denoted in the drawings as the *z*-

direction.

[0019] With particular reference to FIG. 2, the illustrative waveguide **10** has a first portion **12**, a second portion **14**, and a transitional portion **16** with a first end **20** joining the first portion **12** and a second end **22** opposite the first end joining the second portion **14**. The first portion **12** of the waveguide **10** has a smaller thickness (or height) than the second portion **14** of the waveguide **10**, and the transitional portion **16** of the waveguide **10** has monotonically increasing thickness when moving along the +x-direction from its first end **20** to its second end **22**. Notably, there is no abrupt step in the transitional portion **16** of the waveguide **10**.

[0020] With particular reference back to FIG. 1, in the top view there seen the optical structure further includes (in addition to the waveguide **10**) a first structure **30** disposed along a first side of the waveguide **10**, and a second structure **32** disposed along a second side of the waveguide **10** opposite the first side. The first and second structures **30** and **32** are not optically operative components of the optical structure. Rather, as will be described later, the first and second structures **30** and **32** provide for control of a chemical-mechanical polishing (CMP) step of the fabrication of the waveguide **10**. This CMP control provides the gradual increase in thickness of the transitional portion **16** of the waveguide **10**. Hence, the first structure **30** is also referred to herein as a first CMP control structure **30**; and likewise the second structure **32** is also referred to herein as a second CMP control structure **32**.

[0021] FIGS. 1 and 2 further illustrate a light emitter **34** disposed at an input end **36** of the first portion **12** of the waveguide **10**. As seen in FIG. 2, the input end **36** of the first portion **12** is distal from the transitional portion **16** of the waveguide **10**. The light emitter **34** may, for example, be an output end of an optical fiber, or a laser, light emitting diode (LED), or the like. In operation, the waveguide **10** of the illustrative optical structure of FIGS. 1 and 2 receives light L output by the light emitter **34** into the input end **36** of the first portion of the waveguide **10**. In a typical waveguide, there would be an abrupt step in height (i.e., a thickness step) between the lower-height first portion **12** and the higher-height second portion **14**; and the abrupt step in height would introduce light reflection that reduces light transmission efficiency of the waveguide. In the illustrative waveguide **10**, however, the gradual increase in thickness of the transitional portion **16** advantageously reduces or eliminates such reflections and substantially improves the optical efficiency of the waveguide **10**, and in particular of light transmission from the first portion **12** through the transitional portion **16** to the second portion **14**.

[0022] The illustrative transitional portion **16** of the waveguide **10** has a changing thickness which is linearly increasing from the first portion **12** of the waveguide **10** to the second portion **14** of the waveguide **10** (or, said another way, is linearly increasing from its first end **20** to its second end **22**). More generally, to limit or eliminate light reflection the transitional portion **16** should have a changing thickness which is monotonically increasing from the first portion **12** of the waveguide **10** to the second portion **14** of the waveguide **10** (or, said another way, the changing thickness is monotonically increasing from its first end **20** to its second end **22**).

[0023] With reference now to FIGS. 3-5, formation of the gradually changing thickness of the transitional portion **16** of the waveguide **10** by way of control of a chemical-mechanical polishing (CMP) step is described. FIG. 3 shows the top view of FIG. 1 with certain dimensions indicated, while FIG. 4 shows the C0-C0 cut view of the waveguide **10** with certain dimensions indicated. Starting with FIG. 4, the first portion **12** of the waveguide **10** has a thickness or height H1, the second portion **14** of the waveguide **10** has a thickness or height H3 (where $H3 > H1$), and the transitional portion **16** of the waveguide **10** has a gradually increasing thickness from its first end **20** to its second end **22**, with a selected thickness H2 indicated in FIG. 2 for a cut line C2-C2 (where $H1 < H2 < H3$). FIG. 4 also indicates a cut line C1-C1 passing through the first portion **12**, and a cut line C3-C3 passing through the second portion **14**. Each cut plane C1-C1, C2-C2, and C3-C3 is a y-z plane using the x-y-z Cartesian directions indicated in the drawings. Note that while FIG. 4 depicts a certain cut line C1-C1 through the first portion **12**, any cut line through the first

portion **12** would have the same thickness **H1** since the first portion **12** has a constant thickness along the x-direction; and likewise, while FIG. **4** depicts a certain cut line **C3-C3** through the second portion **14**, any cut line through the second portion **14** would have the same thickness **H3** since the second portion **14** has a constant thickness along the x-direction. On the other hand, as a hypothetical cut line through the transitional portion **16** moves from its first end **20** to its second end **22** the height would increase, from the height **H1** at the first end **20** to the height **H3** at the second end **22**. The illustrative cut plane **C2-C2** with height **H2** is thus an illustrative example. [0024] Referencing FIG. **3**, the corresponding cut planes **C1**, **C2**, and **C3** passing through the first portion **12**, transitional portion **16**, and second portion **14** respectively, are again indicated. At the cut plane **C1-C1** through the first portion **12**, the first CMP control structure **30** is spaced apart from the waveguide **10** by a spacing **S1** (along the y-direction); and likewise, the second CMP control structure **32** at the cut plane **C1-C1** is spaced apart from the waveguide **10** by the same spacing **S1**. The first CMP control structure **30** has uniform spacing **S1** over its entire length along the x-direction.

[0025] At the cut plane **C3-C3** through the second portion **14**, the first CMP control structure **30** is spaced apart from the waveguide **10** by a smaller spacing **S3**; and likewise, the second CMP control structure **32** at the cut plane **C3-C3** is spaced apart from the waveguide **10** by the same spacing **S3**. The second CMP control structure **32** has uniform spacing **S3** over its entire length along the x-direction.

[0026] On the other hand, along the transitional portion **16** of the waveguide **10** the CMP control structures **30** and **32** have decreasing spacing from the waveguide **10** when moving along the +x-direction from the first end **20** to the second end **22** of the transitional portion **16** of the waveguide **10**. FIG. **3** depicts each of the CMP control structures **30** and **32** at the example cut plane **C2-C2** having a spacing **S2** from the waveguide **10** (where $S1 < S2 < S3$).

[0027] It is noted that to simplify the analysis depicted in FIGS. **3-5**, the waveguide **10** is shown in FIG. **3** has having a uniform width in the y direction, thus neglecting the slight gradual increase in width in the y-direction of the waveguide **10** when moving along the +x-direction as depicted in FIG. **1**.

[0028] With reference now to FIGS. **5A-5H**, a method of fabricating the optical structure including the waveguide **10** and CMP control structures **30** and **32** is illustrated by way of successive diagrammatically shown cut views taken along the cut **C3-C3** indicated in FIGS. **3** and **4** at successive steps of the fabrication process. As labeled by Cartesian direction coordinates in each of FIGS. **5A-5H**, the cut plane of the cut **C3-C3** is a y-z cut plane which is perpendicular to the x-direction. In this illustrative example, the fabrication process starts with a silicon-on-insulator (SOI) wafer **40** as shown in FIG. **5A**. The SOI wafer **40** includes a silicon layer **42** disposed on a buried oxide (BOX) layer **44**. The SOI wafer **40** may, for example, be a commercial SOI wafer. The buried oxide layer **44** is described herein as a silicon dioxide layer as an illustrative example, but other suitable oxide materials are contemplated. The buried oxide layer **44** should have a refractive index which is less than the refractive index of silicon, so that the buried oxide layer **44** can serve as a lower-index cladding material beneath the waveguide **10** destined to be formed from the silicon layer **42**. While an SOI wafer **40** is described as the base substrate for fabricating the optical structure, it will be appreciated that other base substrates can be employed that provide a silicon layer disposed on an oxide layer with refractive index less than that of the silicon wafer. As another nonlimiting illustrative example, the structure could be a silicon substrate on which a suitable oxide layer is deposited followed by deposition of a silicon layer. Moreover, while the illustrative examples employ the silicon layer **42** for forming a silicon waveguide **10** (and silicon CMP control structures **30** and **32**), it is contemplated to employ another waveguide material as the layer **42** so as to form a waveguide of a different material using the fabrication method here described. Thus, more generally, the layer **42** may be considered a waveguide layer **42**, and the layer **44** may be considered a bottom cladding layer **44**.

[0029] With reference to FIG. 5B, the fabrication process starts by deposition of an etch stop layer **46** on the silicon layer **42**. In the illustrative example, an initial thin silicon dioxide (SiO.sub.2) layer **48** is deposited before depositing the etch stop layer **46**, but the initial SiO.sub.2 layer **48** is optional. In the examples herein, the etch stop layer **46** is a silicon nitride (SiN) layer, but other etch stop layer materials are contemplated. The etch stop layer **46** is suitably deposited by chemical vapor deposition (CVD), physical vapor deposition (PVD), sputtering, or another deposition modality.

[0030] With reference to FIG. 5C, photolithographically patterned etching of the etch stop layer **46** is performed to remove the etch stop layer **46** except at locations corresponding to the destined waveguide **10** and CMP control structures **30** and **32**. The photolithographically patterned etching entails disposing photoresist on the etch stop layer **46**, performing a photolithographic exposure of the photoresist to form a latent image in the photoresist, developing the latent image to remove areas exposed to light (if a positive photoresist is used) or to remove areas not exposed to light (if a negative photoresist is used) so as to openings in the photoresist, followed by etching to remove the etch stop layer **46** in the openings of the photoresist to form corresponding openings **50** in the etch stop layer **46**. In the illustrative example of FIG. 5C, the underlying SiO.sub.2 layer **48** serves as an etch stop for the etching of the etch stop layer **46**; however, in other contemplated embodiments if the etchant used to form the openings **50** is selective for silicon nitride (or other material forming the etch stop layer **50**) over the silicon layer **42** (or other waveguide material making up the layer **42**), then the SiO.sub.2 layer **48** may optionally be suitably omitted. In FIG. 5C, the etch stop layer is now labeled as a patterned etch stop layer **46P** due to the etching to form the openings **50**, to indicate the patterning of this layer. Note that while not visible in the cut view along cut C3-C3 shown in FIG. 5C which employs a y-z cut plane, the patterned etch stop layer **46P** if viewed in top view (i.e., along the z-direction) would have a pattern corresponding to the locations of the waveguide **10** and CMP control structures **30** and **32** as depicted in FIGS. 1 and 3.

[0031] With reference to FIG. 5D, the silicon layer **42** is etched through the openings **50** in the patterned etch stop layer **46P** to remove the silicon layer **42** except in the areas underneath the patterned etch stop layer **46P**. The etchant employed is highly selective for etching silicon (or other waveguide material making up the layer **42**) over silicon nitride (or other etch stop material making up the patterned etch stop layer **46P**). The etchant employed is also effective to remove the optional SiO.sub.2 layer **48** except underneath the patterned etch stop layer **46P**, so that this is now a patterned SiO.sub.2 layer **48P** as labeled in FIG. 5D. The silicon which remains underneath the patterned etch stop layer **46P** after the etching forms the waveguide **10** and CMP control structures **30** and **32**. The waveguide **10** and CMP control structures **30** and **32** at the stage diagrammatically depicted in FIG. 5D have the layout shown in FIGS. 1 and 3, but the waveguide **10** does not yet have the thickness (i.e., height) profile shown in FIGS. 2 and 4. Rather, at the stage diagrammatically depicted in FIG. 5D the waveguide **10** has a uniform height which is greater than (or at smallest, equal to) the thickness H3 labeled in FIG. 4 for the second portion **14** of the completed waveguide **10**. The etching leaves the optical structure at the stage diagrammatically depicted in FIG. 5D with spaces **52** between the waveguide **10** and the CMP control structures **30** and **32**.

[0032] With reference to FIG. 5E, the spaces **52** between the waveguide **10** and the CMP control structures **30** and **32** is filled with a cladding material **54**. In the illustrative example, the cladding material **54** is silicon dioxide and hence, after the filling with the cladding material **54**, the illustrative silicon waveguide **10** is surrounded, i.e., encapsulated, by cladding material comprising silicon dioxide. Namely, the waveguide **10** is clad on its bottom by the buried oxide **44** of the original SOI wafer **40** (see FIG. 5A) and on its lateral sides and top by the filling cladding material **54**. Furthermore, since in the illustrative embodiment the buried oxide **44**, the filling cladding material **54**, and the patterned SiO.sub.2 layer **48P** are all silicon dioxide, at the fabrication stage diagrammatically shown in FIG. 5E these silicon dioxide regions form a continuum of silicon

dioxide which encapsulates the waveguide **10**, the CMP control structures **30** and **32**, and the patterned etch stop layer **46P**. In some embodiments, the filling of the spaces **52** between the waveguide **10** and the CMP control structures **30** and **32** with the silicon dioxide cladding material **54** is performed by a shallow trench isolation (STI) process. In a suitable STI process, an initial conformal layer of silicon dioxide is optionally formed by thermal oxidation, followed by deposition of silicon dioxide by CVD or another suitable deposition modality to a thickness sufficient to fill the spaces **52** and extend over and cover the waveguide **10** and CMP control structures **30** and **32** and the overlying patterned etch stop layer **46P**, as diagrammatically shown in FIG. 5E.

[0033] An STI process typically includes a subsequent planarization by chemical-mechanical polishing (CMP). This is described next with reference to FIG. 5F. CMP is sometimes referred to in the art as “chemical-mechanical planarization”. However, in the optical structure fabrication method here described, the CMP does not produce a planar surface. This is because the structure to which CMP is applied has a laterally varying structure imposed by the waveguide **10** and the CMP control structures **30** and **32**. This laterally varying structure, as will be explained later herein with particular reference to FIG. 6, results in a thickness reduction of the silicon waveguide **10** by the CMP that varies in amount depending on the spacing between the waveguide **10** and the CMP control structures **30** and **32** (e.g., spacing **S1** or **S2** or **S3** labeled in FIG. 3). The amount of thickness reduction may also depend on the width of the CMP control structures (i.e., width along the y-direction), as a larger width for the CMP control structures can increase the resistance to the CMP. The thickness reduction may also depend on width of the waveguide **10** (see, e.g., FIG. 1 showing the waveguide width in the y-direction increasing when moving along the +x-direction—a wider waveguide presents more resistance to the CMP and hence will have less thickness reduction).

[0034] With reference to FIG. 5F, the impact of the CMP control structures **30** and **32** is illustrated for the cut C3-C3 through the second portion **14** of the waveguide **10**. (In the C3-C3 cut plane, the CMP control structures **30** and **32** have spacing **S3** away from the waveguide **10**, as labeled in FIG. 3). The CMP control structures **30** and **32** present more resistance to the CMP than does the silicon dioxide cladding **54** disposed between the waveguide **10** and each of the CMP control structures **30** and **32**. This greater resistance is due to the silicon dioxide of the cladding **54** presenting less resistance to the CMP than does the silicon of the CMP control structures **30** and **32**, as well as that the patterned etch stop **46P** disposed on the CMP control structures **30** and **32** is typically more resistant to the CMP than the silicon dioxide of the cladding **54**. As recognized herein, this results in the CMP not providing perfect planarization, but rather producing a diagrammatically shown bowed or dished surface **56**. The bowed or dished surface **56** results from the CMP more efficiently removing material of the cladding **54** and waveguide **10** compared with the CMP control structures **30** and **32** which are more resistant to the CMP processing. Notably, the amount of bowing or dishing of the surface **56** produced by the CMP will depend on the structure of the CMP control structures **30** and **32**, particularly the spacing of these structures away from the waveguide **10**. This will be discussed in greater detail below with reference to FIG. 6.

[0035] In some embodiments, the method of fabricating the optical structure including the waveguide **10** may terminate at the structure shown in FIG. 5F.

[0036] With reference to FIG. 5G, in other embodiments an optional further process operation of depositing additional silicon dioxide (or, more generally, additional cladding material) **58** may be performed. This may be beneficial if, for example, the CMP performed to produce the structure as shown in FIGURE#5F entirely removes the silicon dioxide cladding from the top surface of the waveguide **10** (see the examples for cut C1-C1 and cut C2-C2 shown in FIG. 6). In this case, the additional deposited silicon dioxide **58** ensures that the waveguide **10** is entirely embedded in (i.e., encompassed by) the (illustrative silicon dioxide) cladding material with lower refractive index than that of the (illustrative silicon) of the waveguide **10**, so that light is guided through the

waveguide **10** by total internal reflection (TIR).

[0037] With reference to FIG. 5H, in some embodiments an optional second CMP process may be performed after depositing the additional silicon dioxide **58**, to provide a planarized surface **60** for the final optical structure. Because this second CMP operates only on a uniform layer of silicon dioxide (or other cladding material, in other embodiments) and does not penetrate deeply enough to expose the CMP control structures **30** and **32**, this second CMP is expected to produce a more planar final surface (as opposed to the bowed or dished surface **56** produced by the first CMP due to interaction with the CMP control structures **30** and **32**).

[0038] The fabrication method has been described with reference to FIGS. 5A-5H which diagrammatically depict successive cut views at the cut plane C3-C3 through the second portion **14** of the waveguide **10**. These views do not illustrate how the gradually changing thickness along the x-direction of the transitional portion **16** of the waveguide **10** is obtained.

[0039] With reference back to FIGS. 3 and 4, and with further reference now to FIGS. 6-8, it is explained how the CMP control structures **30** and **32** control the CMP to provide the height profile of the waveguide **10** shown in FIG. 4, and in the particular example how the gradual increase in thickness of the transitional portion **16** from H1 to H3 when moving along the +x-direction is obtained. FIG. 6 diagrammatically illustrates cut views for cut C1-C1; FIG. 7 diagrammatically illustrates cut views for cut C2-C2; and FIG. 8 diagrammatically illustrates cut views for cut C3-C3. Each of FIGS. 6, 7, and 8 depicts two cut views: one presenting the structure before the CMP processing (as previously described with reference to FIG. 5F), and one presenting the structure after the CMP process. To clarify, each of FIGS. 6, 7, and 8 include an arrow (->) labeled "CMP" to clarify the before/after cut views. The lefthand (before-CMP) cut view of FIG. 6 for cut C1-C1 also labels the spacings S1 of the first and second CMP control structures **30** and **32** at the cut C1-C1, which are also labeled in FIG. 3. The lefthand (before-CMP) cut view of FIG. 7 for cut C2-C2 also labels the spacings S2 of the first and second CMP control structures **30** and **32** at the cut C2-C2, which are also labeled in FIG. 3. The lefthand (before-CMP) cut view of FIG. 8 for cut C3-C3 also labels the spacings S3 of the first and second CMP control structures **30** and **32** at the cut C3-C3, which are also labeled in FIG. 3. Furthermore, for simplicity the silicon dioxide cladding is labeled in the aggregate as cladding **64** (as opposed to separately labeling the cladding components **44**, **48P**, and **54** as was done in the successive cut views of FIGS. 6A-6H). As previously discussed, $S1 > S2 > S3$, as also seen by comparing the lefthand (before-CMP) drawings of FIGS. 6, 7, and 8.

[0040] As previously discussed when describing the CMP processing with reference to FIG. 5F, The CMP control structures **30** and **32** present more resistance to the CMP than does the silicon dioxide cladding **54** disposed between the waveguide **10** and each of the CMP control structures **30** and **32**, due to the silicon dioxide cladding **54** presenting less resistance than the silicon of the CMP control structures **30** and **32**, and due to the patterned etch stop **46P** disposed on the CMP control structures **30** and **32** being more resistant to the CMP than the silicon dioxide cladding **54**. The amount of resistance to the CMP is also dependent on the width of the waveguide **10**. In the example of FIG. 1, the waveguide **10** has a width (along the y-direction) which increases with increasing distance along the +x direction, so that the waveguide **10** itself also presents increasing resistance to the CMP with increasing distance along the +x direction. These effects produces the bowed or dished surface **56** after the CMP as indicated in FIG. 5F. It is further noted that the amount of thickness reduction may also depend on the width of the CMP control structures (i.e., width along the y-direction), as a larger width for the CMP control structures can increase the resistance to the CMP.

[0041] The amount of the bowing or dishing is expected to increase with increasing spacing between the waveguide **10** and the CMP control structures **30** and **32**. This spacing is S1 for cut C1-C1 of FIG. 6, and is a smaller spacing S2 for the cut C2-C2 of FIG. 7, and a still smaller spacing S3 for the cut C3-C3 of FIG. 8. That is, $S1 > S2 > S3$. Consequently, the bowing or dishing should be greatest for the largest spacing S1 of FIG. 6, and should be least for the smallest spacing

S3 of FIG. 8, and should be intermediate for the intermediate spacing S2 of FIG. 7. As seen in FIGS. 3 and 4, in the transitional portion 16 of the waveguide 10 the spacing between the waveguide 10 and the CMP control structures 30 and 32 decreases in continuous fashion from the largest spacing S1 at the end 20 of the transitional portion 16 to the smallest spacing S3 at the opposite end 22 of the transitional portion 16, with the spacing S2 being at an intermediate point between the ends 20 and 22. This decreasing spacing is expected to produce a gradual reduction in the bowing or dishing produced by the CMP along the x-direction from end 20 to end 22 of the transitional portion 16, and hence the thickness reduction produced by the CMP is expected to be gradually less and less when moving from the end 20 to the end 22. This produces the desired gradually increasing thickness (or height) of the transitional portion 16 of the waveguide 10. This effect is illustrated in FIGS. 6, 7, and 8 by the labeled bowed or dished surface 561 for cut C1-C1 shown in FIG. 6; which has greater bowing than the labeled bowed or dished surface 562 for cut C2-C2 shown in FIG. 7; which has greater bowing than the labeled bowed or dished surface 563 for cut C3-C3 shown in FIG. 8.

[0042] In addition to changes in bowing or dishing due to the different spacings $S1 > S2 > S3$, the total amount of material removed is expected to depend on the overall resistance imposed by the waveguide 10 itself. This depends on the width of the waveguide 10. In the example of FIG. 1, the width of the transitional portion 16 of the waveguide 10 gradually increases from the first end 20 of the transitional portion 16 of the waveguide 10 to the second end 22 of the transitional portion 16 of the waveguide 10.

[0043] In general, the slope or curvature of the thickness of the transitional portion 16 of the waveguide 10 will depend on the spacing between the waveguide 10 and the CMP control structures 30, 32, and may also depend on other factors such as the width of the waveguide 10 itself. To design the waveguide 10 with a transitional portion 16 having a thickness variation along its length with a particular desired slope or curvature, a test matrix of test structures can be fabricated with different spacing variations (e.g., using different test structures with layouts corresponding to FIG. 3 but with different values for S1, S2, and/or S3 in each test structure) and optionally also different waveguide widths (e.g., as shown in FIG. 1), and CMP processing can be performed on these test structures, followed by characterization of the thickness slope or curvature obtained for the transitional region 16 in each test case using a technique such as cross-sectional microscopy, profilometry, or the like to determine an optimal geometry for the optical structure to obtain the desired thickness slope or curvature for the transitional region 16 of the waveguide 10.

[0044] With reference back to FIGS. 1 and 2, in that example the transitional portion 16 was used to minimize or eliminate optical reflection of light L input to the input end 36 of the first portion 12 by the light emitter 34, thereby increasing the optical efficiency of the waveguide 10. It will be appreciated that this is just one of many possible applications of waveguides with the illustrative gradual transition that does not include any light-reflecting thickness (i.e., height) steps.

[0045] With reference to FIG. 9, another nonlimiting illustrative application example is shown. The example of FIG. 9 employs the same waveguide 10 including the first portion 12, second portion 14, and interposed transitional portion 16 with first end 20 connecting with the first portion 12 and second end 22 connecting with the second portion 14. However, the application depicted in FIG. 9 is different. In FIG. 9, the waveguide 10 is delivering light L to a photodetector 70, such as a photodiode or the like. In this case, the second portion 14 of the waveguide has a larger thickness (i.e., height) to accommodate a larger size of the photodetector 70. The transitional portion 16 in this application provides for transmission of the light L from the lower-thickness first portion 12 to the larger-thickness second portion 14 with reduced or eliminated light reflection. Again, it is to be understood that the light input coupling application shown in FIGS. 1 and 2 as well as the light output coupling application shown in FIG. 9 are nonlimiting illustrative applications, and other applications are contemplated such as providing couplings from a waveguide of smaller height to a waveguide of greater height with reduced or eliminated light reflections.

[0046] In the following, some further embodiments are described.

[0047] In a nonlimiting illustrative embodiment, a method of fabricating a waveguide is disclosed. The method includes: forming a stack of layers including an etch stop layer disposed on a waveguide layer disposed on a bottom cladding layer; patterning the etch stop layer and etching the waveguide layer after the patterning to form a waveguide and a chemical-mechanical polishing (CMP) control structure; filling a space between the waveguide and the CMP control structure with a cladding material; and performing CMP to reduce a thickness of the waveguide, wherein the CMP control structure controls the CMP of the waveguide to form a transitional portion of the waveguide having a gradually changing thickness.

[0048] In a nonlimiting illustrative embodiment, an optical structure comprises a waveguide having a first portion, a second portion, and a transitional portion with a first end joining the first portion and a second end joining the second portion. A thickness of the first portion of the waveguide is smaller than a thickness of the second portion of the waveguide. The transitional portion of the waveguide has a thickness that gradually increases from the first thickness at the first end of the transitional portion of the waveguide to the second thickness at the second end of the transitional portion of the waveguide.

[0049] In a nonlimiting illustrative embodiment, a method of fabricating a waveguide having a variable thickness formed by chemical-mechanical polishing (CMP) is disclosed. The method includes: depositing an etch stop layer on a silicon layer of a silicon-on-insulator (SOI) wafer; patterning the etch stop layer and etching the silicon layer after the patterning to form a first CMP control structure, a second CMP control structure, and a silicon waveguide disposed between the first CMP control structure and the second CMP control structure; filling spaces between the silicon waveguide and the first and second CMP control structures with silicon dioxide using a shallow trench isolation (STI) process; and performing CMP to reduce a thickness of the silicon waveguide, wherein the first and second CMP control structures control the CMP of the silicon waveguide to form a transitional portion of the silicon waveguide having a gradually changing thickness.

[0050] In a nonlimiting illustrative embodiment, a waveguide has a first and second portions, and a transitional portion with a first end joining the first portion and a second end joining the second portion. The first portion has a first thickness that is smaller than a second thickness of the second portion. The transitional portion has a thickness that gradually increases from the first thickness at its first end to the second thickness at its second end. In a fabrication method employing chemical-mechanical polishing (CMP), first and second CMP control structures are disposed on opposite sides of the waveguide. Spaces between the waveguide and the CMP control structures are filled with cladding material. CMP is performed to reduce a thickness of the waveguide. The CMP control structures control the CMP of the waveguide to form the transitional portion of the waveguide having the gradually increasing thickness.

[0051] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A waveguide fabrication method comprising: forming a stack of layers including an etch stop layer disposed on a waveguide layer disposed on a bottom cladding layer; patterning the etch stop layer and etching the waveguide layer after the patterning to form a waveguide and a chemical-

- mechanical polishing (CMP) control structure; filling a space between the waveguide and the CMP control structure with a cladding material; and performing CMP to reduce a thickness of the waveguide, wherein the CMP control structure controls the CMP of the waveguide to form a transitional portion of the waveguide having a gradually changing thickness.
2. The method of claim 1, wherein the CMP control structure includes: a first CMP control structure disposed along a first side of the waveguide and spaced apart from the waveguide by a spacing that gradually changes over a portion of the first CMP control structure disposed along the transitional portion of the waveguide; and a second CMP control structure disposed along a second side of the waveguide opposite the first side of the waveguide and spaced apart from the waveguide by a spacing that gradually changes over a portion of the second CMP control structure disposed along the transitional portion of the waveguide; wherein the gradually changing spacings of the first and second CMP control structures disposed along the transitional portion of the waveguide controls the CMP of the waveguide to produce the gradually changing thickness of the transitional portion of the waveguide.
 3. The method of claim 1, wherein the waveguide has a width that gradually increases over the transitional portion of the waveguide.
 4. The method of claim 1, wherein the waveguide layer comprises silicon and the cladding material comprises silicon dioxide.
 5. The method of claim 4, wherein the etch stop layer comprises silicon nitride.
 6. The method of claim 4, wherein the space between the waveguide and the CMP control structure is filled with the cladding material using a shallow trench isolation (STI) process.
 7. The method of claim 1, wherein the CMP control structure controls the CMP of the waveguide to produce the transitional portion of the waveguide having the changing thickness which is monotonically increasing from a first portion of the waveguide to a second portion of the waveguide, wherein after performing the CMP a thickness of the first portion of the waveguide is smaller than a thickness of the second portion of the waveguide.
 8. The method of claim 7, further comprising one of: disposing a light emitter at an input end of the first portion of the waveguide wherein the input end is distal from the transitional portion of the waveguide; or disposing a light detector at an output end of the main portion of the waveguide wherein the input end is distal from the transitional portion of the waveguide
 9. The method of claim 7, wherein after performing the CMP the transitional portion of the waveguide does not include an abrupt thickness step.
 10. The method of claim 1, wherein the forming of the stack of layers includes: depositing the etch stop layer on a silicon-on-insulator (SOI) wafer; wherein the waveguide layer of the stack of layers comprises a silicon layer of the SOI wafer and the cladding layer of the stack of layers comprises a buried oxide layer of the SOI wafer.
 11. An optical structure comprising: a waveguide having a first portion, a second portion, and a transitional portion with a first end joining the first portion and a second end joining the second portion; wherein a thickness of the first portion of the waveguide is smaller than a thickness of the second portion of the waveguide; and wherein the transitional portion of the waveguide has a thickness that gradually increases from the first thickness at the first end of the transitional portion of the waveguide to the second thickness at the second end of the transitional portion of the waveguide.
 12. The optical structure of claim 11, wherein the waveguide comprises silicon.
 13. The optical structure of claim 12, further comprising: a first structure comprising silicon disposed along a first side of the waveguide and spaced apart from the waveguide by a spacing that gradually changes over a portion of the first structure disposed along the transitional portion of the waveguide; and a second structure comprising silicon disposed along a second side of the waveguide opposite the first side of the waveguide and spaced apart from the waveguide by a spacing that gradually changes over a portion of the second structure disposed along the transitional

portion of the waveguide.

14. The optical structure of claim 12, wherein a width of the transitional portion of the waveguide gradually increases from the first end of the transitional portion of the waveguide to the second end of the transitional portion of the waveguide.

15. The optical structure of claim 12, further comprising: a cladding comprising silicon dioxide surrounding the waveguide at least on a bottom and sides of the waveguide.

16. The optical structure of claim 11, wherein the transitional portion of the waveguide does not have any abrupt thickness step.

17. A method of fabricating a waveguide having a variable thickness formed by chemical-mechanical polishing (CMP), the method comprising: depositing an etch stop layer on a silicon layer of a silicon-on-insulator (SOI) wafer; patterning the etch stop layer and etching the silicon layer after the patterning to form a first CMP control structure, a second CMP control structure, and a silicon waveguide disposed between the first CMP control structure and the second CMP control structure; filling spaces between the silicon waveguide and the first and second CMP control structures with silicon dioxide using a shallow trench isolation (STI) process; and performing CMP to reduce a thickness of the silicon waveguide, wherein the first and second CMP control structures control the CMP of the silicon waveguide to form a transitional portion of the silicon waveguide having a gradually changing thickness.

18. The method of claim 17, wherein: a spacing between the first CMP control structure and the silicon waveguide gradually changes over a portion of the first CMP control structure disposed along the transitional portion of the silicon waveguide; and a spacing between the second CMP control structure and the silicon waveguide gradually changes over a portion of the second CMP control structure disposed along the transitional portion of the silicon waveguide; wherein the gradually changing spacing of the first CMP control structure and the gradually changing spacing of second CMP control structure controls the CMP of the silicon waveguide to produce the gradually changing thickness of the transitional portion of the silicon waveguide.

19. The method of claim 17, wherein the etch stop layer comprises silicon nitride.

20. The method of claim 17, wherein the filling of the spaces between the silicon waveguide and the first and second CMP control structures comprises performing a shallow trench isolation (STI) process to fill the spaces between the silicon waveguide and the first and second CMP control structures.
